

Summary Data on Acid Cropland and Lime Requirements in Tanzania

GAIA

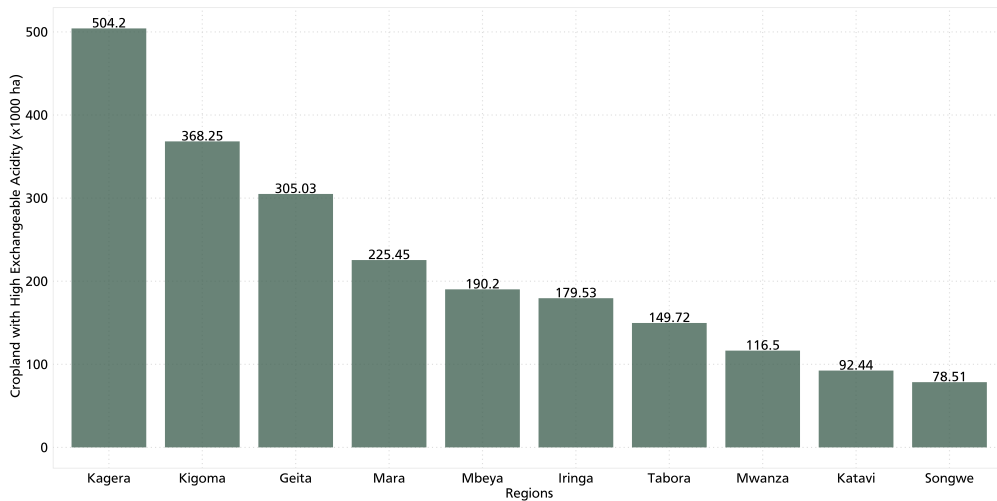
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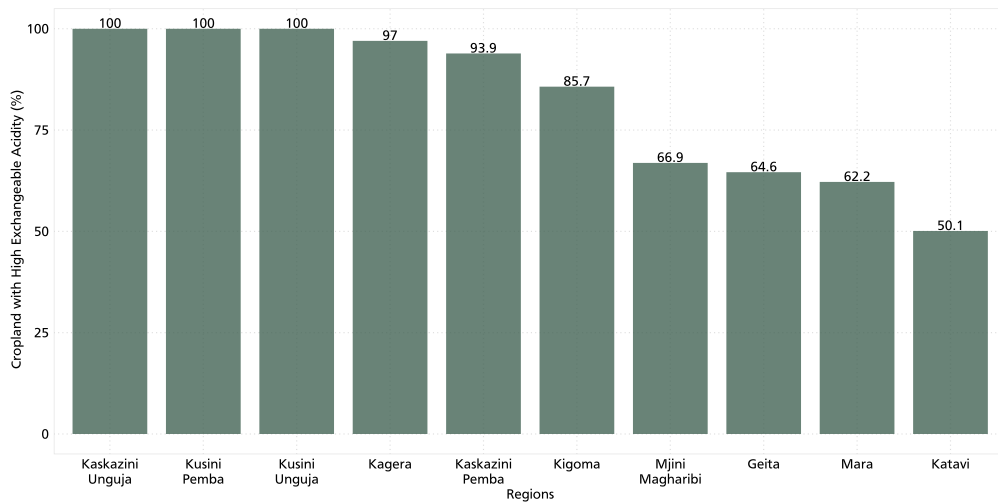
Outlook

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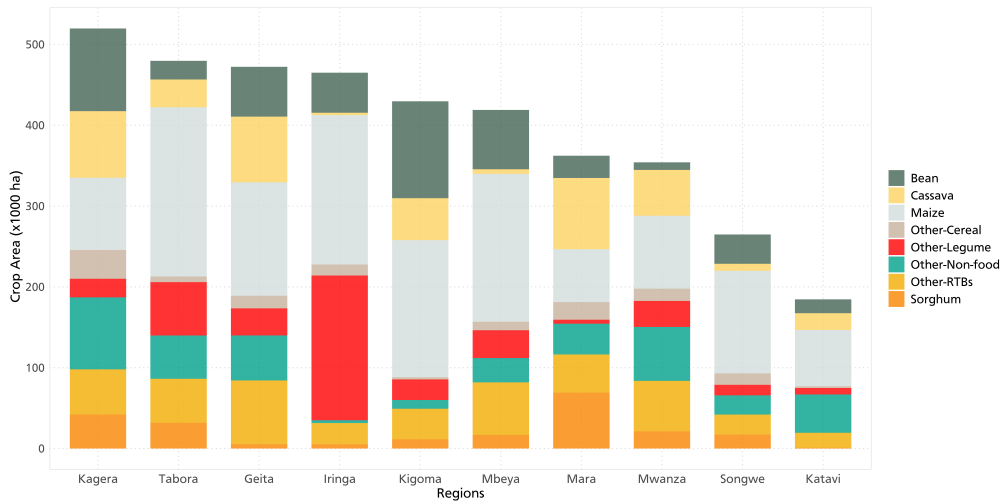
Top 10 Regions in Tanzania with the highest acidic cropland



Top 10 Regions in Tanzania with the highest acidic cropland (Percentage of total cropland)



Total crop area in the top-10 most acidic Regions, not segregated by acidic cropland



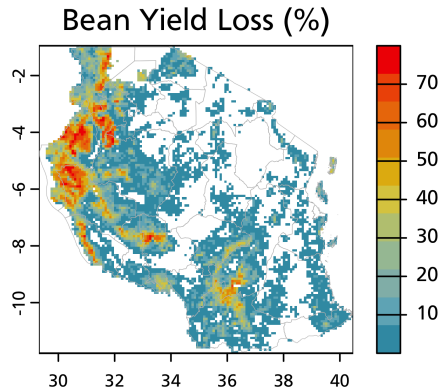
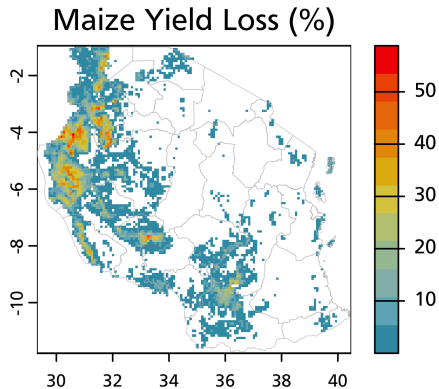
Tanzania-Summary

└ Total crop area distribution (ha)

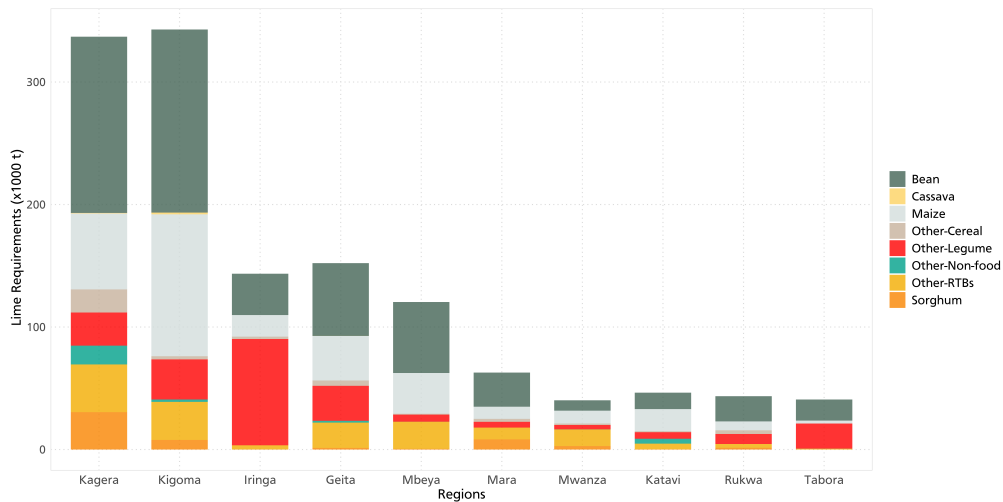
└ Total crop area in the top-10 most acidic Regions, not segregated by acidic cropland

Regions	Bean	Cassava	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Sorghum
Geita	61.62	81.38	140.31	15.57	33.46	55.71	79.03	5.21
Iringa	49.55	2.95	184.81	13.70	179.01	3.50	26.34	5.18
Kagera	102.26	82.45	89.30	35.70	22.96	89.08	55.95	42.02
Katavi	17.08	20.77	69.70	2.14	8.02	47.37	18.60	0.89
Kigoma	119.90	51.83	169.76	2.62	25.52	10.77	37.86	11.38
Mara	27.52	88.24	65.43	21.82	4.85	38.03	47.33	69.10
Mbeya	73.37	5.62	183.11	10.51	34.35	30.19	65.01	16.78
Mwanza	9.36	56.74	90.11	15.23	32.22	66.68	62.60	21.13
Songwe	36.22	8.65	126.95	14.15	13.00	23.79	24.90	17.15
Tabora	23.04	34.40	209.44	6.93	66.03	53.56	54.51	31.78

Yield loss due to soil acidity for Maize and Bean in Tanzania



Lime requirements (t) for acid soil remediation in Tanzania

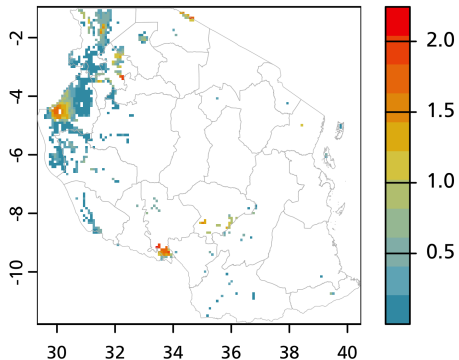


Lime requirements (t) for acid soil remediation in Tanzania (table)

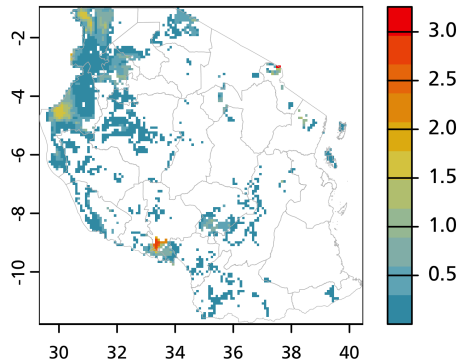
Regions	Bean	Cassava	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Sorghum
Geita	59.42	0.06	36.32	4.35	28.57	1.48	20.71	1.27
Iringa	33.75	-	17.61	1.93	86.78	-	3.29	0.16
Kagera	143.91	0.37	62.01	18.90	27.00	15.39	38.95	30.48
Katavi	13.40	0.02	18.01	0.83	5.39	3.96	4.58	0.23
Kigoma	149.50	1.38	115.89	2.58	32.72	1.95	31.09	7.79
Mara	27.83	-	9.93	2.42	4.69	-	9.68	8.25
Mbeya	57.90	-	33.32	0.68	5.79	0.01	22.11	0.58
Mwanza	8.48	-	10.55	1.17	3.53	0.10	13.60	2.79
Rukwa	20.58	-	7.28	3.01	8.17	-	3.14	1.36
Tabora	17.22	-	2.34	0.05	20.46	0.06	0.65	0.05

Lime rate for Maize and Bean in Tanzania

Maize Lime Rate (t/ha)

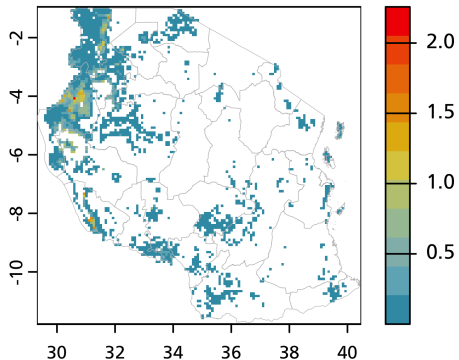


Bean Lime Rate (t/ha)

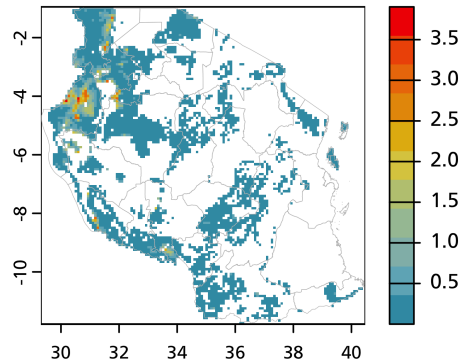


Yield response to lime application in Tanzania

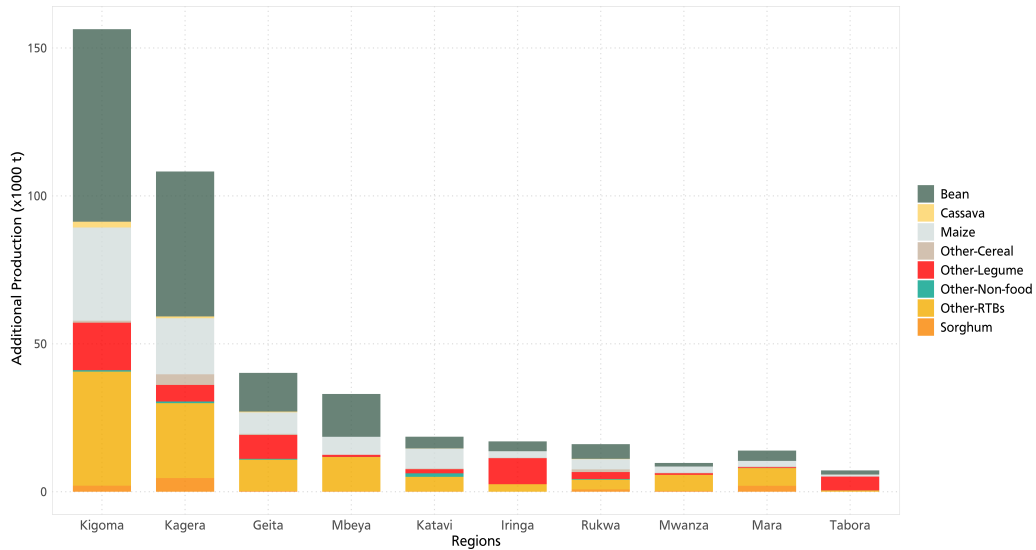
Maize Yield Response (t/ha)



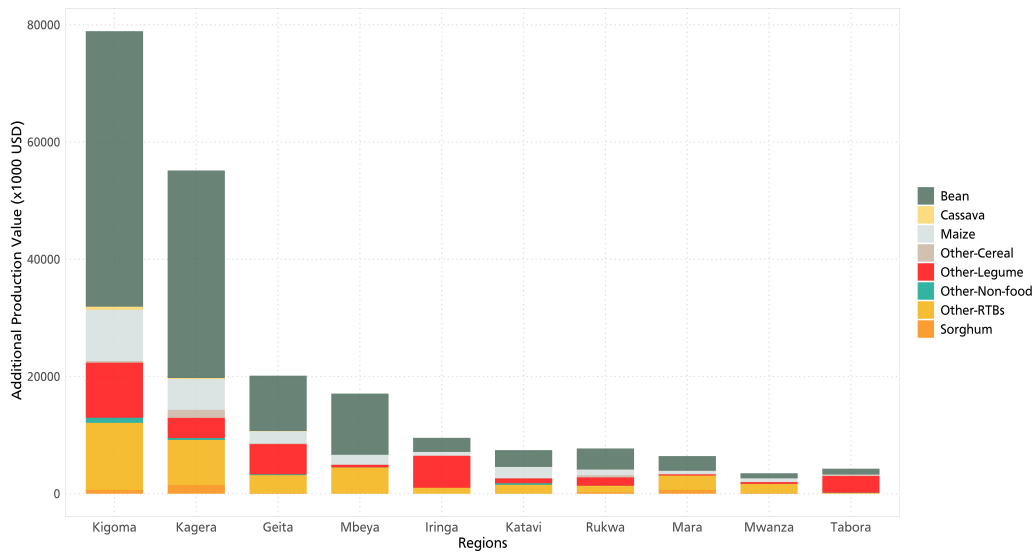
Bean Yield Response (t/ha)



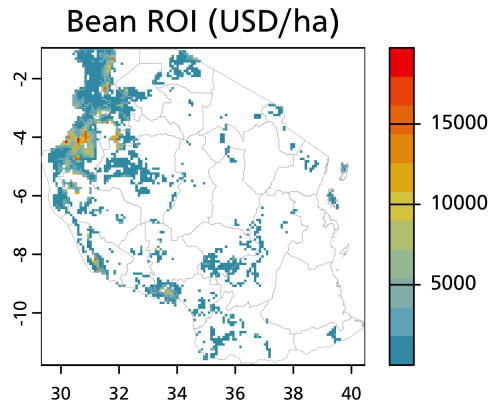
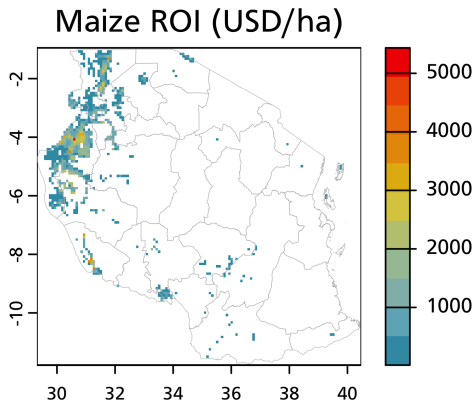
Potential for additional production (tone) in top 10 acidic Regions in Tanzania



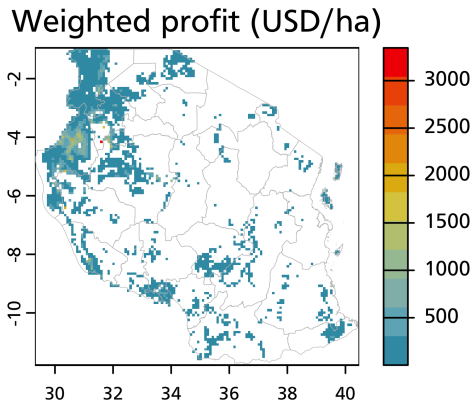
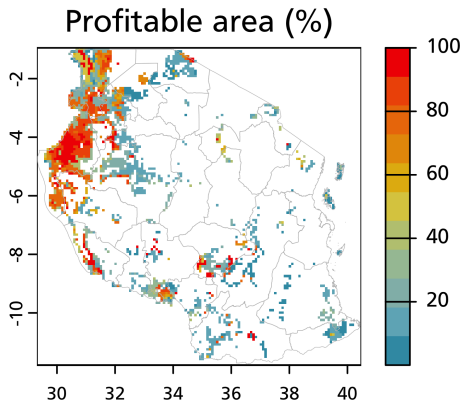
Potential for additional production value (US\$) in top 10 acidic Regions in Tanzania



Return on investment for lime application in Tanzania



Profitability of lime application in Tanzania



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

For details, visit www.acidsoils.africa.